Forest Ecology and Management 453 (2019) 117633

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

Blowdown disturbance effect on the density, richness and species composition of the seed bank in Central Amazonia

Natali Gomes Bordon[a,*], Anselmo Nogueira[b], Niwton Leal Filho[c], Niro Higuchi[d]

[a] Programa de Pós-Graduação em Botânica, Instituto Nacional de Pesquisas da Amazônia, Av. André Araújo, 2936, Aleixo, Manaus, Amazonas 69060-001, Brazil
[b] Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, Rua Arcturus, Jardim Antares, São Bernardo do Campo, São Paulo 09606-070, Brazil
[c] Departamento de Ecologia, Instituto Nacional de Pesquisas da Amazônia, Av. André Araújo, 2936, Petrópolis, C.P. 478, Manaus, Amazonas 69060-001, Brazil
[d] Departamento de Silvicultura Tropical, Manejo Florestal, Instituto Nacional de Pesquisas da Amazônia, Av. André Araújo, 2936, Petrópolis, C.P. 478, Manaus, Amazonas 69067-375, Brazil

ARTICLE INFO

Keywords:
Forest disturbance
Melastomataceae
Large gaps
Araceae
Tree fall

ABSTRACT

Windstorms, known as blowdowns, create large canopy gaps in the tropical rainforest. Despite the occurrence of blowdowns in Central Amazonia, no studies have yet investigated the seed bank in areas altered by blowdown in the Amazon rainforest. The seed bank plays an important role in the regeneration of larger natural gaps, and changes in, or partial loss of, seed bank can modify the successional dynamics locally. In the present study, we compared the density, richness and floristic composition of a seed bank in a post-blowdown secondary forest with approximately six years old and a continuous forest. The seed bank was characterized by collecting 160 soil samples systematically distributed in each forest type at two different times. Samples were collected at a minimum distance of 30 m from each other, sampling the superficial soil 5 cm in depth and 20 cm in diameter. At each sampling point, the slope and canopy openness was also measured. The density and floristic composition of the seed bank was estimated using counts of seedlings that had emerged in a nursery of the collected soil samples. The post-blowdown secondary forest exhibited seed banks 30% denser and 20% richer, on average, than continuous forest. The floristic composition also differed between forest types, with a predominance of Melastomataceae, Urticaceae, and Araceae in the secondary forest, followed by Cyperaceae and Moraceae; however, these did not differ between forest types. In contrast, for other families, greater richness and density were observed in the continuous forest. Higher density and richness of trees, shrubs, epiphytes, and hemi-epiphytes were observed in the secondary forest, except for the richness of epiphytes. In general, terrain slope was unrelated to density, richness and floristic composition of the seed bank. Under a disturbance of great magnitude, such as a blowdown, the density and richness of the seed bank were not reduced; on the contrary, its potential for local regeneration was maintained. Despite the limited contribution of the seed bank to the regeneration of climax species, the seed bank was shown to be diverse in species, including different life forms, and may contribute to the regeneration of forest strata, even after severe disturbances.

1. Introduction

The temporal and spatial variation of the seed bank in tropical forests is influenced mainly by local plant community, historical land use, and forest matrix (Dalling and Denslow, 1998; Dupuy and Chazdon, 1998; Garwood, 1989; Miranda et al., 2009). Spatial variation of the seed bank occurs both in horizontal and vertical directions, in which the upper layers of soil have the highest diaspore abundance (Harper, 1977; Young et al., 1987). Temporal variation of the seed bank is determined by rates of loss and incorporation of the seeds into the soil (Garwood, 1989; Harper, 1977). Early successional species contribute with higher densities to the seed bank, and, generally, seeds have high longevity, while late successional species contribute with lower densities, and the seeds have low longevity (Baider et al., 2001; Baskin and Baskin, 1998; Garwood, 1989; Lindner, 2009). Consequently, the floristic composition of seed bank does not fully reflect the species composition from the standing vegetation of old forests (Cui et al., 2016; Hopfensperger, 2007). Therefore, the function of the seed bank is to preserve diaspores of a set of species that generally occur in low density in the forest canopy of old forests (Baskin and Baskin, 1998; Garwood, 1989; Harper, 1977). These diaspores are critical at the beginning of forest regeneration after the occurrence of disturbances (Castillo and

* Corresponding author.
E-mail addresses: natalibordon@yahoo.com.br, n.libordon@gmail.com (N.G. Bordon).

https://doi.org/10.1016/j.foreco.2019.117633
Received 5 July 2019; Received in revised form 11 September 2019; Accepted 14 September 2019
Available online 14 October 2019
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Stevenson, 2010; Dalling and Hubbell, 2002; Metcalfe and Turner, 1998; Saulei and Swaine, 1988).

Forest dynamics is regulated by the frequency and intensity of disturbances (Brokaw, 1985; Sousa, 1984). The formation of canopy gaps has a direct impact on the seed bank, initially decreasing seed density very rapidly in the soil owing to the increased rate of seedling emergence (Garwood, 1989; Harper, 1977; Young et al., 1987). Then, the subsequent establishment of the early successional species leads to an increase of seed input in soil, and with the advance of forest succession, seed density decreases gradually (Martins and Engel, 2007; Quintana-Ascencio et al., 1996; Saulei and Swaine, 1988; Young et al., 1987). The size of canopy gaps in tropical forests is the main factor controlling the contribution of the seed bank for forest regeneration, given that large gaps increase the local and surrounding amplitude of light and temperature (Brown, 1996; Denslow, 1980, 1987; Pearson et al., 2002, 2003).

The formation of large gaps in the tropical rainforests can be caused by windstorms known as blowdowns (Chambers et al., 2009; Garstang et al., 1998), which are high velocity descending air currents that are generally associated with intense rainfall in tropical areas (Garstang et al., 1998; Hjelmfelt, 2007). Blowdowns impact forests in different ways, compromising broad extensions of the forest canopy (Marra et al., 2014; Nelson et al., 1994). Blowdowns smaller than 50 ha are more often but blowdowns > 101 ha, although uncommon, account for more than half of the disturbed areas (Espírito-Santo et al., 2010), and they can suppress the tree cover (Marra et al., 2014). In particular, they overthrow trees and other plants, breaking trunks and branches, uprooting trees and causing defoliation, generally increasing dead plant biomass and exposing soil horizons (Boose et al., 2004; Brokaw and Walker, 1991; Marra et al., 2014; Rifai et al., 2016). In addition, the type of disturbance leaves affected areas susceptible to recurrence of further falls of trees and branches (Brokaw and Walker, 1991; Uriarte et al., 2005), which can lead to depletion of the seed bank.

In tropical rainforests, the seed bank accounts for the great number of regenerants in large natural gaps (Brown, 1996; Castillo and Stevenson, 2010; Denslow, 1980; Pearson et al., 2002, 2003) like those caused by blowdowns. In those cases, the persistence of the seed bank would increase plant community resilience after disturbances (Alexander et al., 2012; Auffret et al., 2015; Díaz-Perea et al., 2014). However, partial loss of the seed bank could modify the process of ecological succession, e.g., in areas under extreme anthropogenic impact (Arroyo-Rodríguez et al., 2017; Chua et al., 2016; Mesquita et al., 2015). Specifically, the Amazon rainforest landscape has traditionally been converted to pasture after burning, which has led to the elimination of most seed banks, followed by the domination of early successional *Vismia* species during regeneration (Longworth et al., 2014; Mesquita et al., 2001, 2015). In contrast, unburnt areas, partly maintaining their seed banks, were initially occupied by *Cecropia* species, which favor subsequent regeneration of a broader set of plant species under their canopies (Jakovac et al., 2015; Mesquita et al., 2001, 2015; Monaco et al., 2003; Williamson et al., 2014).

Although the occurrence of blowdowns in the Amazon rainforest has been reported (Chambers et al., 2009; Garstang et al., 1998; Marra et al., 2014; Negrón-Juárez et al., 2010; Nelson et al., 1994; Rifai et al., 2016), studies evaluating the impact of blowdowns on the seed bank are not known in the literature. Given that the seed bank is the main propagule source in plant regeneration after the formation of large natural gaps in tropical rainforest (Castillo and Stevenson, 2010; Ellison et al., 1993; Pearson et al., 2002; Pearson et al., 2003; Turner et al., 1998), it becomes crucial to characterize the seed bank available for forest regeneration under these conditions, especially in cases of such dramatic disturbance events as blowdowns. Also, the terrain slope may intensify plant damage during the occurrence of disturbances (Bellingham, 1991; Marra et al., 2014; Rifai et al., 2016), consequently modifying the dynamic of the seed bank and plant regeneration.

In order to investigate the effects of blowdown and topographic variation of the seed bank, we compared the density, richness, and composition of the seed bank in two different ecological conditions: a secondary forest approximately six years old after a blowdown event, and a continuous forest without evidence of disturbance. We hypothesized that regeneration after blowdown alters the density and richness of the seed bank, modifying, in turn, species composition, when compared to the continuous forest, and that steeper areas intensify seed bank modification after blowdown, but not in continuous forest. More specifically, large disturbances make the affected forest more susceptible to new events of treefalls and branches (e.g. Brokaw and Walker, 1991; Marra et al., 2014; Uriarte et al., 2005), which could cause a general seed bank depletion due to the high recruitment rate and subsequent mortality. On the other hand, these disturbed areas offer a wide variation of regeneration microenvironments (e. g. Clinton and Baker, 2000; Comita et al., 2009; Guzmán-Grajales and Walker, 1991; Putz, 1983; Shiels et al., 2010; Walker, 2000; Walker et al., 2003) that could favor a higher abundance and richness of pioneer species in the seed bank. If so, forest regeneration after large disturbances could favor trees, shrub, and lianas pioneer species, in addition to increasing herbaceous species.

2. Material and methods

2.1. Study area

This study was carried out in the northern region of Brazil at the Experimental Station of Tropical Forestry (EEST) belonging to the National Institute of Amazonian Research (INPA) and adjacent-contiguous area with access by road ZF2, which is administered by the Superintendence of the Manaus Free Zone (SUFRAMA) (Higuchi et al., 1998), located about 90 km northwest of Manaus. Both areas are characterized by lowland evergreen “terra firme” in tropical rainforest with closed canopy 30–37 m high covering the landscape with significant edaphic-topographic variation (Higuchi et al., 1998; Ribeiro et al., 1999). In the flat and higher areas, the soils are more clayey, and in the alluvial plains along streams, sandier soils predominate (Ferraz et al., 1998). The climate of the region is warm and humid with an annual average rainfall between 1750 and 2500 mm and an average annual temperature of around 27 °C (Peel et al., 2007; Ribeiro, 1976). The rainy season extends from November to May, after which rainfall declines, August and September then being the driest months (Ribeiro, 1976).

To investigate the variation of the density and diversity of the seed bank, we performed a study case comparison between two adjacent areas under different disturbance conditions: i) secondary forest, approximately six years old, altered after blowdown event (Fig. 1A-D) (2°56′22.43″ S, 60°26′72.94″ W) (Marra et al., 2014) and ii) continuous forest without evidence of forest disturbances (Fig. 1E and F) (2°61′39.35″ S, 60°20′35.03″ W) (Higuchi et al., 1998). Both secondary and continuous areas have a forest matrix without evidence of anthropic activities such as pasture and agriculture. The blowdown event in the secondary forest area occurred on January 2005 with downburst velocities between 26 and 41 ms^{-1} associated with heavy rainfall, impacting a total of 2668 ha of tropical rainforest with an estimated mortality of 0.32 ± 0.05 million of trees in the Manaus region (Negrón-Juárez et al., 2010). The post-blowdown area was chosen because it had the most severe disturbance pattern throughout the landscape, losing > 13% of the large trees by sub-plot for more details see Marra et al. (2014).

2.2. Seed bank sampling

To sample seed banks in both areas, we used pre-established trails by Marra et al. (2014) and Higuchi et al. (1998). The sampling process was systematized, by following continuous trails throughout the landscape. The seed bank samples were collected at a minimum distance of



Fig. 1. Areas sampled in the Experimental Station of Tropical Forestry (EEST) of the National Institute of Amazonian Research (INPA) and adjacent-contiguous area with access by road ZF2: A) clearing in reconstruction in the post-blowdown secondary forest, B) canopy under reconstruction in the post-blowdown secondary forest, C) secondary forest with logs and uprooting of trees, D) logs and organic matter in soil of secondary forest, E) aspect of continuous forest understory in plateau area, F) slope of continuous forest.

30 m from each other. In each area, we performed 80 soil samples in the summer and 80 in the winter, totaling 160 soil samples (N_{total} = 320 samples). The seed bank samplings were standardized using a template (PVC pipe) 20 cm in diameter and 5 cm in height to collect superficial soil free from the thick litter.

2.3. Slope and canopy opening sampling

In each sample, we performed local measurements of slope and canopy opening. The slope was measured using a clinometer (Sunnto) and a fixed beacon in the direction of the greater slope of the terrain at a distance of 10 m from the observer. The canopy opening was estimated by means of a spherical crown densitometer (Lemmon, 1957) at 1.3 m of soil height.

2.4. Emerged seedlings from the seed bank: Density, diversity, and life forms

We used an indirect method to quantify the density and floristic composition of the seed bank, which consisted of counting seedlings that had emerged in a nursery (Brown, 1992; Gross, 1990). Soil samples were scattered in trays approximately 0.5 cm deep (Dalling et al., 1994). The trays were filled with washed sand and sawdust (3:1) and randomly distributed in the greenhouse covered by a screen with 50% shading. The seedlings were counted from the seed banks at two day intervals, and in the fourth and fifth months, seedlings were counted on a weekly basis because of the reduced speed of seedling emergence. To control for the potential occurrence of contaminants in the greenhouse, ten trays containing only washed sand and sawdust were also installed. In these trays, some seedlings were easily characterized as contaminants and were excluded from density and floristic estimates in all trays. The seedlings that emerged from the seed bank were morphotyped, and at least four specimens were transplanted and allowed to grow out in order to facilitate species identification. Plant species identification was made after consulting the relevant seedling literature (Camargo et al., 2008; Duke, 1965; Garwood, 2009; Vogel, 1980) for family (Gentry, 1996; Simpson, 2006), genus (Berg, 1978; Berg et al.,

Table 1
Slope and canopy opening of post-blowdown secondary forest and continuous forest and density, including number of seedlings and richness of the seed bank (mean ± standard deviation).

Variables	Secondary forest	Continuous forest
Slope (%)	12.9 ± 8.9	14.6 ± 13.4
Canopy opening (%)	6.8 ± 2.2	5.7 ± 2.1
Number of seedlings per sample	22 ± 24	16 ± 14
Density of seedlings (m^2)	704 ± 770	498 ± 437
The overall number of seedlings	3540	2502
Seedlings identified at the family level	3171	2217
Seedlings identified at the morphospecies level	2363	1779
Overall number of species	86	90
Individuals/species of pioneers seedlings (%)	59/67	55/62
Individuals/species of no-pioneers seedlings (%)	10/33	19/38
Seedlings not classified in succession groups (%)	31/−	26/−

1984; Michelangeli, 2005), flora (Berry et al., 2006; Ribeiro et al., 1999; Steyermark et al., 2001) and by comparison of specimens with exsiccates from the Herbarium of the National Institute of Amazonian Research (INPA) and with the aid of taxonomists, parataxonomists and experts.

We estimated richness and floristic composition considering only individuals identified at the species or morphospecies level, including species grouping, in particular *Bellucia* spp. and Rubiaceae spp., which correspond to two or more species that were not recognized in our samples. Based on the literature and information in the exsiccates, seedlings were also categorized according to the life forms, such as trees, shrubs, herbs, lianas, epiphytes and hemiepiphytes (Ribeiro et al., 1999). In some cases, life forms were inferred by growth pattern and stem consistency. We also categorized species from the seed bank into pioneer and non-pioneer species based on the classification proposed by Swaine and Whitmore (1988) with a few modifications.

2.5. Statistical analyses

In order to test the effect of forest type – secondary forest after blowdown versus continuous forest – and slope variation on the density and diversity of seed bank, we used generalized linear mixed models (GLMM), in which variance was not constant, and errors were not normally distributed (Crawley, 2007). Canopy opening was highly correlated with forest type, in which secondary forest was, on average, 1.1° higher than continuous forest ($z = -5.02$, $p < 0.01$, N = 320). Thus, canopy opening was not included in our final models to explain seed bank variation.

In the GLMMs, we used different quantitative descriptors of seed bank as a response variable, forest type (categorical), and slope variation (continuous) as fixed factors, and sampling season as a random variable. Since density and richness of seed bank were counting descriptors in our models, we used Poisson or Negative Binomial error distribution. Since soil samples with more seedlings are more likely to contain more species, we included seedling density as a covariate in all models exploring species richness. In all models, we inspected the overdispersion parameter of residuals before interpreting significance (Crawley, 2007) and biological consequences.

Plant species composition was explored using two different analytical approaches, the first by reducing the dimensionally of the plant community using multivariate ordination techniques and the second after exploring patterns of the five most abundant plant families individually. Among the 6042 emerged seedlings, around 90% were identified at the family level, 70% were identified at species or morphospecies level, and the rest of the seedlings died before they could be identified. In general, non-identified seedlings were composed of seedlings from abundant families occurring in a large number of samples. Additionally, the most common families in our seed bank samples (e.g., Melastomataceae, Urticaceae, and Araceae) were identified with confidence by taxonomists based on their unique morphologies. The dimensionality of plant species composition was reduced using hybrid multidimensional scaling analysis (nMDS). We carried out two ordinations with plant dataset: one based on quantitative data and another using presence-absence data. We used the Bray-Curtis distance measure on site-standardized (percentage of each species in each site) data in quantitative ordinations with the objective of capturing the patterns displayed by the most abundant species that had the greatest quantitative contribution to the differences among samples. We used the Sorensen distance index in the presence-absence ordination to capture patterns generated by rarer species. The resulting nMDS ordination axes, which describe the community composition, were used in graphical representations (nMDS axes 1 and 2) and as response variables in general linear mixed models (LMM) testing effects of forest type and slope variation on plant composition.

Finally, we categorized the species and morphospecies sampled in the seed bank in trees, shrubs, herbs, lianas, epiphytes, and hemiepiphytes. For each life form, we tested the effect of forest type and slope variation on seed bank density and richness using GLMMs similar to analyses described before.

3. Results

In total, 6042 seedlings were recorded, considering all seed bank samples, in which 3540 seedlings emerged from the secondary forest and 2502 from the continuous forest. We identified a total of 121 species, considering all samples in which 86 species were sampled from the secondary forest, and 90 from the continuous forest (Table 1). The variation in slope of the continuous forest was more pronounced than that in secondary forest (Table 1), but did not differ between forest types ($F = 1.75$, $p = 0.19$, N = 320), while the canopy opening had lower variation, but did differ between forest types ($F = 21.12$, $p < 0.01$, N = 320).

3.1. Differences of seed bank density and richness between forest types and slope gradient

Secondary forest samples had, on average, 0.29 more seedlings than continuous forest ($z = -3.88$, $p < 0.01$, N = 320) (Fig. 2A) and had, on average, 0.21 more species than continuous forest ($z = -4.52$, $p < 0.01$, N = 320) (Fig. 2B). In contrast, seedling density ($z = 0.03$, $p = 0.97$, N = 320) (Fig. 2A) and species richness ($z = -1.19$, $p = 0.23$, N = 320) (Fig. 2B) were unrelated to the slope gradient.

3.2. Differences of seed bank composition between forest types and slope gradient

3.2.1. Species composition described by nMDS ordination

Quantitative ordination of species composition captured 57% of the variation exhibited by the original distances between seed bank samples in three dimensions (nMDS axis 1, 2 and 3) (Fig. 3A). The first ($z = 17.50$, $p < 0.01$, N = 319) ordination axis was related to forest type, while the second ($z = 1.18$, $p = 0.24$, N = 319) and third ($z = -0.12$, $p = 0.90$, N = 319) axes were not related. Ordination of presence-absence data captured 42% of the variation exhibited by the original distances between samples in five dimensions (Fig. 3B). Similar to the quantitative ordination, the first ordination axis ($z = 16.20$, $p < 0.01$, N = 319) was related to forest type, while the second ($z = -1.81$, $p = 0.07$, N = 319), third ($z = -1.54$, $p = 0.12$, N = 319), fourth ($z = 1.61$, $p = 0.11$, N = 319) and fifth ($z = 1.14$, $p = 0.26$, N = 319) axes were not related.

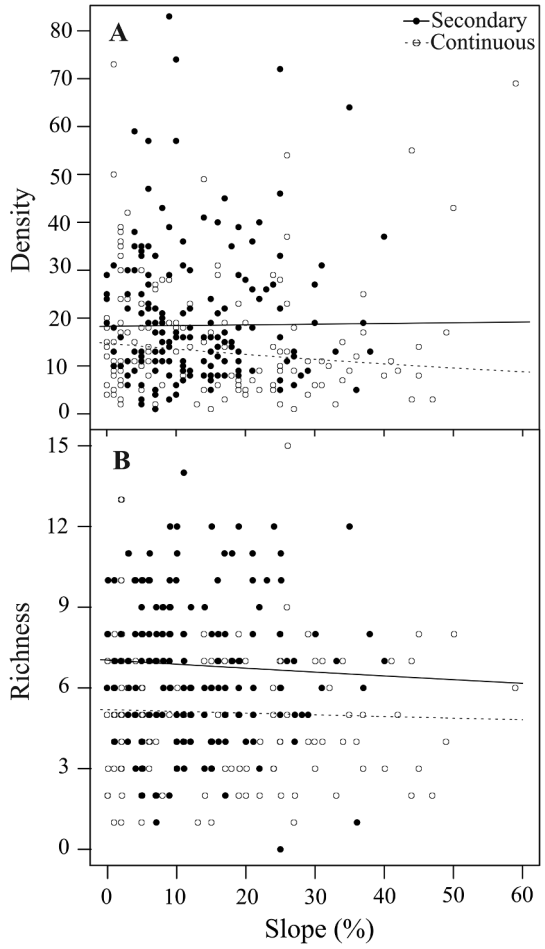


Fig. 2. Slope variation on density per sample (A) and richness per sample (B) of seed bank of post-blowdown secondary forest and continuous forest. The fitted line shows the fixed effect of forest type in generalized linear mixed models. The sample is 20 cm in diameter by 5 cm in height. In each forest type, the total surface area of the samples collected was 5.03 m^2.

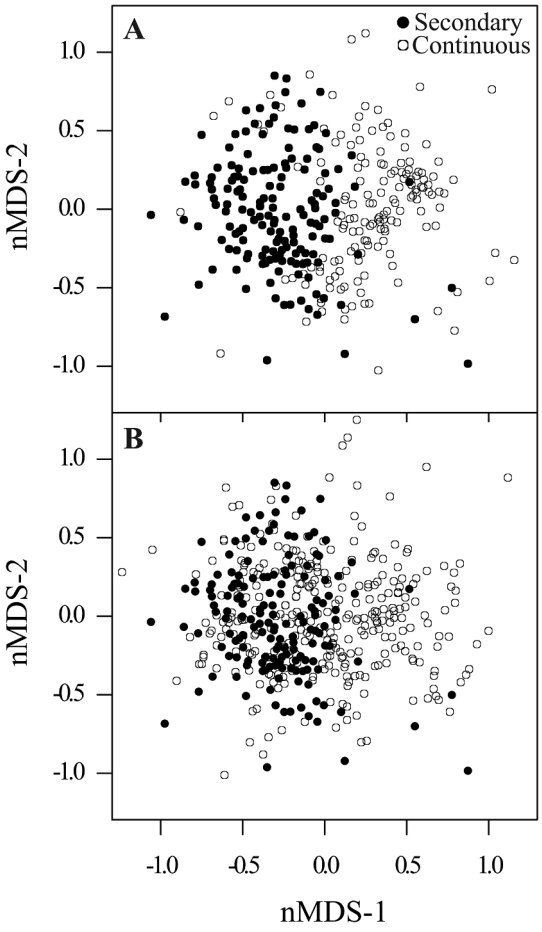


Fig. 3. Ordination diagrams from hybrid multidimensional scaling analysis (nMDS) based on quantitative data using the Bray-Curtis distance (A) and presence-absence of species using the Sorensen distance index (B) of floristic composition of the seed bank of post-blowdown secondary forest and continuous forest. The sample is 20 cm in diameter by 5 cm in height. In each forest type, the total surface area of the samples collected was 5.03 m^2.

3.2.2. Density and richness variation of the most common plant families

Melastomataceae, Urticaceae and Araceae were the most important families in the seed bank in both forest types (Supplementary material, Table S1). Melastomataceae was the most abundant and species-rich group in the seed bank, corresponding to 60% of seedlings that emerged from the post-blowdown secondary forest and 55% that emerged from the continuous forest. Secondary forest samples had a higher density ($z = -3.13$, $p < 0.01$, N = 320) (Fig. 4A) and higher richness ($z = -5.03$, $p < 0.01$, N = 320) (Fig. 4B) of Melastomataceae than samples from continuous forest. The diversity of Melastomataceae was distributed in ten genera: *Miconia* (17 species), *Clidemia* (four species), *Tococa* (three species) and *Bellucia* (two species). *Aciotis*, *Adelobotrys*, *Henriettea, Henriettella, Leandra* and *Maieta* were composed of one species each (Supplementary material, Table S2).

The second most abundant family in the seed bank samples was Urticaceae, corresponding to 10% of the seedlings that emerged from the seed bank of the secondary forest and 9% that emerged from continuous forest (Supplementary material, Table S1). Secondary forest samples had higher density ($z = -4.34$, $p < 0.01$, N = 320) (Fig. 4A) and higher richness ($z = -2.94$, $p < 0.01$, N = 320) (Fig. 4B) of Urticaceae than samples from continuous forest. Urticaceae diversity was distributed in two genera: *Cecropia* (4 species) and *Coussapoa* (3 species) (Supplementary material, Table S2).

The third most abundant family was Araceae, corresponding to 8% of the seedlings that emerged from the secondary forest and 5% that emerged from continuous forest (Supplementary material, Table S1). On average, samples of the secondary forest had higher density ($z = -4.63$, $p < 0.01$, N = 320) (Fig. 4A) and higher richness ($z = -2.73$, $p < 0.01$, N = 320) (Fig. 4B) of Araceae than samples from continuous forest. Araceae diversity was distributed in two genera: *Philodendron* (7 species) and *Heteropsis* (1 species) (Supplementary material, Table S2). Cyperaceae and Moraceae did not differ in the density ($z = -1.15$, $p = 0.25$, N = 320; $z = -1.40$, $p = 0.16$, N = 320; respectively) (Fig. 4A) and richness (Fig. 4B) between forest types.

The remaining plant families had a lower density ($z = 3.25$, $p < 0.01$, N = 320) (Fig. 4A) and richness ($z = 2.3$, $p = 0.02$, N = 320) (Fig. 4B) in secondary forest when compared to continuous forest. For example, Hypericaceae (*Vismia* spp.), Rubiaceae and Goupiaceae (*Goupia glabra*) were more abundant in continuous forest (Supplementary material, Table S1 and S2).

3.2.3. Variation of density and richness per life form between forest types

Samples from secondary forest had, on average, 0.21 more trees ($z = -2.60$, $p < 0.01$, N = 320), 0.76 more shrubs ($z = -4.53$, $p < 0.01$, N = 320), 1.16 more epiphytes ($z = -2.61$, $p = 0.01$, N = 320) and 0.46 more hemiepiphytes ($z = -3.23$, $p < 0.01$, N = 320) than continuous forest (Fig. 5A). No differences in density of herbaceous plants ($z = -0.71$, $p = 0.48$, N = 320) and lianescents ($z = -0.31$, $p = 0.76$, N = 320) were observed between forest types

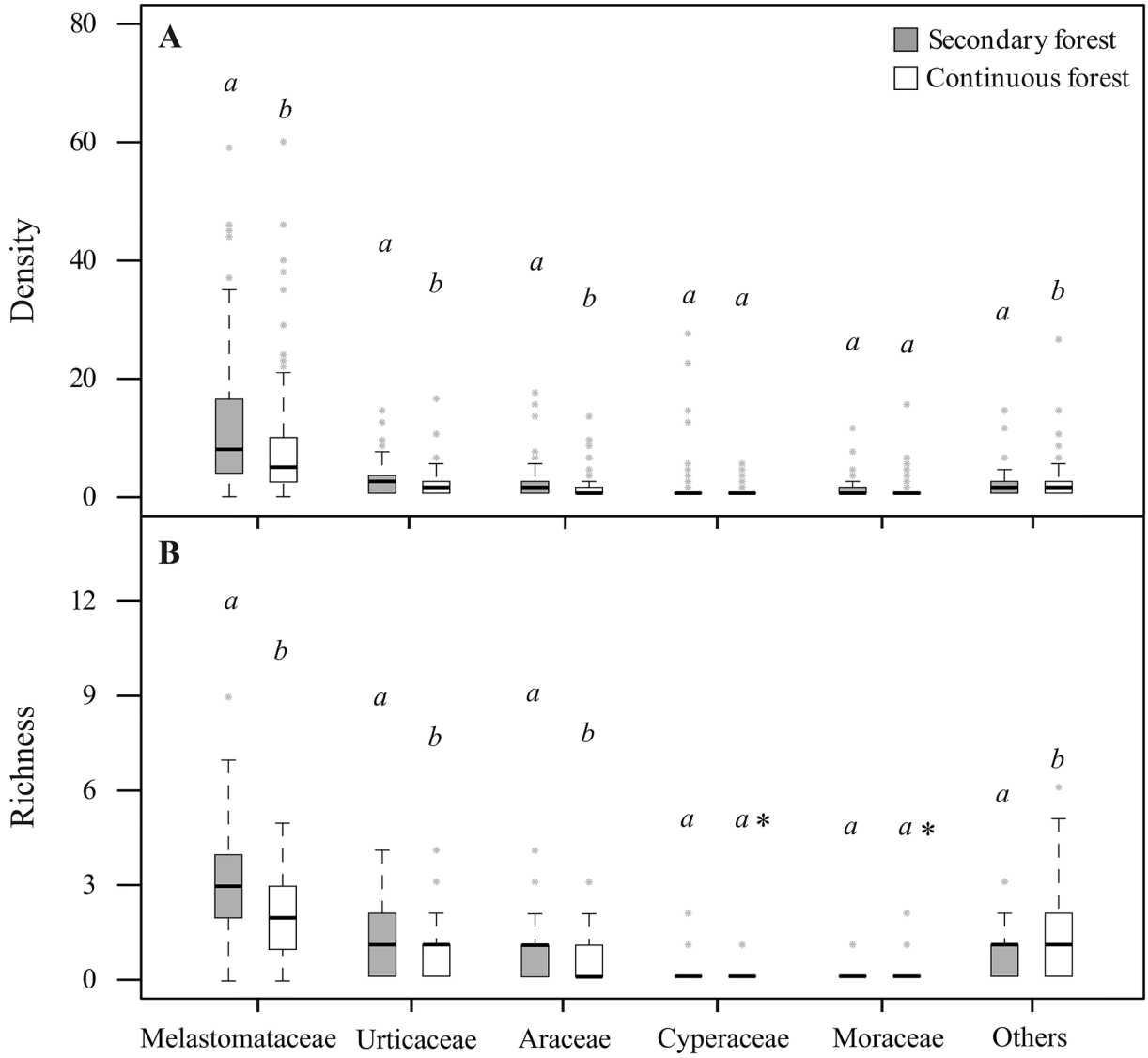


Fig. 4. Variation in density per sample (A) and richness (B) of the five most abundant plant families and other families of the seed bank of the post-blowdown secondary forest and continuous forest. The sample is 20 cm in diameter by 5 cm in height. In each forest type, the total surface area of the samples collected was 5.03 m^2. * Plant families in which GLMM was prohibited because the excess of zeros in the response variable.

(Fig. 5A).

Samples from secondary forest had, on average, 0.25 more trees species ($z = -4.32$, $p < 0.01$, N = 320), 0.54 more shrub species ($z = -3.50$, $p < 0.01$, N = 320) and 0.36 more species of hemiepiphytes ($z = -2.49$, $p = 0.01$, N = 320) than continuous forest (Fig. 5B). No differences in the richness of herbs ($z = -0.94$, $p = 0.35$, N = 320), lianas ($z = 1.21$, $p = 0.23$, N = 320) and epiphytes ($z = -0.88$, $p = 0.38$, N = 320) were observed between forest types (Fig. 5B). Trees have had high density and richness in seed bank, and seedlings of herbs, lianas, epiphytes and hemiepiphytes contributed more to diversity than to density (Table 1). Seed bank was composed mainly of seedlings of pioneer species in both forest types (on average 57% of seedlings and 65% of the species) (Table 1).

4. Discussion

Increase in the frequency of blowdowns may be related to climatic changes (Negrón-Juarez et al., 2010), but little is known about their effect on the seed bank. This is the first study to investigate the impact of blowdown on the seed bank in the Amazon rainforest. As expected in our initial hypotheses, we detected that the seed bank post-blowdown had a higher density and species richness compared to the continuous forest. These differences led to a change in the composition of the seed bank species between forest types, increasing the relative importance of Melastomataceae, Urticaceae, and Araceae in a post-blowdown secondary forest. In a complementary way, variation in the distribution of life forms among forest types, was noted. Here, the post-blowdown secondary forest had a higher density and richness of trees, shrubs, epiphytes and hemiepiphytes, except for the richness of epiphytes. Contrary to our hypotheses, we had no evidence that declivity intensifies the blowdown effect on the seed bank. In sum, our results accord with the pattern of higher seed density in regeneration areas of tropical forests owing to the initial proliferation of pioneer species (Alvarez-Aquino et al., 2005; Garwood, 1989; Lindner, 2009; Miranda et al., 2009; Sousa et al., 2017). Pioneer species generally produce a large number of small seeds with high longevity, which favors their permanence in the seed bank over time (Bentos et al., 2008; Dalling and Hubbell, 2002; Dalling et al., 1998; Garwood, 1989).

In our study, we verified a denser seed bank in secondary forest compared to continuous forest. The seed density of the post-blowdown secondary forest (704 seedlings·m^{-2}) was lower than that found in other secondary forests with different disturbance histories and forest

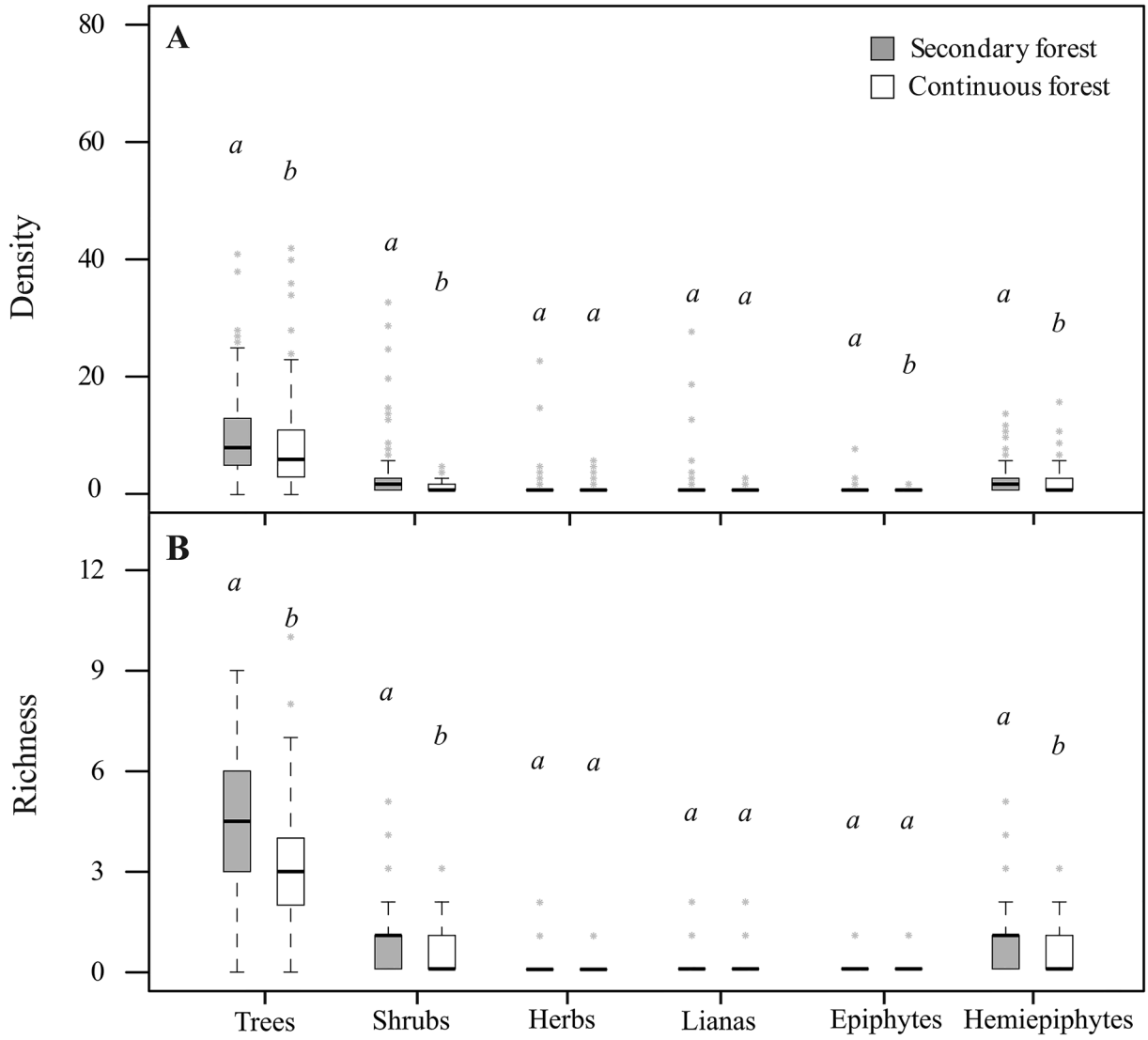


Fig. 5. Variation in density per sample (A) and richness (B) by life forms of the seed bank of the post-blowdown secondary forest and continuous forest. The sample is 20 cm in diameter by 5 cm in height. In each forest type, the total surface area of the samples collected was 5.03 m^2.

matrix (2848 seedlings·m^{-2}, Araujo et al., 2001; 8085 seedlings·m^{-2}, Monaco et al., 2003; 1309 and 1690 seedlings·m^{-2}, Sousa et al., 2017) and in forest fragments and tropical secondary forests (873 and 3632 seedlings·m^{-2}, Alvarez-Aquino et al., 2005; 1144 and 1250 seedlings·m^{-2}, Lindner, 2009; 1740, 2927 and 4893 seedlings·m^{-2}, Miranda et al., 2009; 1576 seedlings·m^{-2}, Wijdeven and Kuzee, 2000). Seed density in soil samples has great spatial variation (Butler and Chazdon, 1998; Dalling et al.,1998; Plue et al., 2010), and areas that undergo blowdown events may increase the spatial heterogeneity of the seed bank. In our study, the seed bank in the secondary forest included samples with different levels of disturbance. Samples less impacted by the blowdown may have contributed to a relatively lower seed density in post-blowdown areas compared to the seed bank from other secondary forests. In contrast, a preserved forest area close to the present study had seed density (388 and 534 seedlings·m^{-2}, Leal Filho et al., 2013; 662 seedlings·m^{-2}, Sousa et al., 2017) similar to that found for continuous forest (498 seedlings·m^{--2}).

Several topographical descriptors, including terrain slope, have been linked to floristic composition and forest dynamics (Costa, 2006; Costa et al., 2005; Luizão et al., 2004; Tuomisto et al., 2016; Vormisto et al., 2000) and could modify the effects of disturbances, such as blowdowns, on the seed bank. However, we did not detect differences in seed density between the steeper and flatter areas. A similar result was found in a tropical rainforest in Central Amazonia with high variation in seed density between samples (Bentos et al., 2013, 2017), which can apparently be explained by processes other than those related to the topography variation.

The high species richness in secondary forest mainly results from a particular set of species, usually the pioneer species, which benefit from increased luminosity (Pearson et al., 2002, 2003; Uriarte et al., 2005). In our study, secondary forest had greater species richness than continuous forest, even when we controlled the effect of seed density. The seed bank in secondary rainforests and forest fragments generally has higher species richness compared to non-altered areas (Araujo et al., 2001; Quintana-Ascencio et al., 1996; Sousa et al., 2017; Young et al., 1987). The type of disturbance may also play a key role in the regeneration response of the plant community (Chambers et al., 2009; Jakovac et al., 2015; Mesquita et al., 2015; Sousa, 1984; Turner et al., 1998). For example, the diversity of understory species was elevated in a subalpine forest after blowdown events when compared to the forest with no evidence of disturbances; however, in post-blowdown areas with selective removal of wood, richness and diversity were lower (del Rio, 2006).

In addition, differences in floristic composition were observed between the two forests with an increase in the relative importance of Melastomataceae, Urticaceae, and Araceae species in the secondary

forest. Forest disturbances increase environmental heterogeneity and generate different regeneration microenvironments (Comita et al., 2009; Guzmán-Grajales and Walker, 1991; Marra et al., 2014; Putz, 1983; Rifai et al., 2016; Walker, 2000; Walker et al., 2003), which may favor species associated with habitats with greater availability of light and nutrients (Clinton and Baker, 2000; Guzmán-Grajales and Walker, 1991; Peterson and Pickett, 1995; Shiels et al., 2010).

In our study, Melastomataceae seedlings generally predominated in the seed bank, as in other Amazonian sites (Bentos et al., 2013; Leal Filho et al., 2013; Sousa et al., 2017). We detected higher density and richness in the post-blowdown secondary forest compared to the continuous forest. In general, Melastomataceae species produce a large number of small seeds with high longevity (Bentos et al., 2008; Carreira and Zaidan, 2007; Ellison et al., 1993; Silveira et al., 2012, 2013), which favors their high density and permanence in the seed bank. Similarly, we had higher density and richness of Urticaceae and Araceae in the secondary forest seed bank, compared to continuous forest. *Cecropia* (Urticaceae) species are recognized as pioneers of large natural canopy gaps (Chambers et al., 2009; Mesquita et al., 2001; Williamson et al., 2014), and as such, they easily establish on post-disturbance debris by having stilt roots (Berg, 1978) that could favor their increase post-blowdown. The Araceae family is very abundant and diverse in tropical rainforests (Ding et al., 2016; Irume et al., 2013; Küper et al., 2004; Nieder et al., 2001), but its presence in seed banks is limited (Dalling and Denslow, 1998; Leal Filho et al., 2013; Metcalfe and Turner, 1998; Quintana-Ascencio et al., 1996; Wijdeven and Kuzee, 2000). Little is known about the role of Araceae species in seed bank and forest succession, even though some species have benefited by increased light availability (Croat, 1988; Hietz-Seifert et al., 1996; Krömer and Gradstein, 2003; Obermüller et al., 2012), which could explain their higher density and richness in the post-blowdown secondary forest. In contrast, the rarest families in our seed bank samples had higher density and richness in the continuous forest. These plants corresponded generally to shade-tolerant species (climax species) characterized by low seed bank densities in tropical rainforests (Alvarez-Aquino et al., 2005; Leal Filho, 2013; Sousa et al., 2017).

In the present study, the post-blowdown regeneration is promoting higher densities and richness of trees, shrubs, epiphytes, and hemiepiphytes (except epiphyte richness). This pattern seems opposite to the observed in forest fragments or secondary forests after anthropogenic disturbances, characterized by the higher density and richness of herbs, especially the presence of invasive herbs species which causes the progressive loss of the seed bank functionality (Alvarez-Aquino et al., 2005; Dupuy and Chazdon, 1998; Gioria et al., 2014; Gioria and Pyšek, 2016; Miranda et al., 2009; Quintana-Ascencio et al., 1996; Silva and Matos, 2006; Wijdeven and Kuzee, 2000). In contrast, the high density of tree seedlings (on average 49%) in comparison to other life forms corresponded in most commonly to the pioneer species in the present study, which is predominant in the seed banks of Amazon rainforest, as well as pioneer shrubs (Araujo et al., 2001; Leal Filho et al., 2013; Sousa et al., 2017). In general, our results highlighted that the seed bank had a great diversity of life forms with the potential to contribute to the restructuring of forest strata, especially in post-disturbance areas, with species belonging to the upper and lower strata of the canopy, as well as the forest understory. This pattern was similar to that described for other forest sites (Hall and Swaine, 1980; Hopkins and Graham, 1983; Leal Filho et al., 2013; Young et al., 1987) where the seed bank included a broad array of life forms, although the importance of species of shrubs, epiphytes and hemiepiphytes is less studied in successional processes in tropical forests.

5. Conclusions

Even in the face of some disturbance of great magnitude and extension, such as blowdown, our results showed no substantial loss of seed bank. On the contrary, the post-blowdown secondary forest had a highly diverse seed bank with a large number of propagules, unlike other sites where anthropic impacts led to the complete suppression of the seed bank, changing the local successional trajectory (Mesquita et al., 2001; Williamson et al., 2014). We showed that the seed bank is fundamentally important to regeneration, resilience and diversity in tropical rainforests. The seedling richness of Melastomataceae and Araceae found in the present study was not previously reported in studies with seed banks in the Amazon; nonetheless, it is necessary to include these groups in forest management, conservation and succession studies to better understand the importance of these groups. In the future, it is expected that the recovery of native tropical forests will be even more important than today, and in this context, it will be possible that the regeneration process may have propagules from groups less considered in studies so far, such as Melastomataceae. In practice, these groups may be managed directly from the seed bank or more appropriately by ex situ cultivation before its use in the recovery of these areas.

Declaration of Competing Interest

None

Acknowledgments

This work was part of the MSc dissertation of the first author developed at the National Institute of Amazonian Research (INPA), produced in the Graduate Program in Botany, with financial support from the National Council for Scientific and Technological Development (CNPq) (grant number 132967/2010-3). We thank the field and logistical support provided by the Experimental Station of Tropical Forestry (EEST) and Forest Management Laboratory (LMF), both belonging to INPA. I would like to thank José Ramos, Tony Vizcarra Bentos, Kátia Gigliola Cangani, Maria de Lourdes da C.S. Soares, Júlia Araújo, and Antonio Carlos Webber for assistance in the identification of botanical material. I would especially like to thank the researchers Daniel Marra, Alberto Vicentini and Veridiana Vizoni Scudeller for suggestions and lessons during the project. Thanks to Laura Lucas Trujillo, Sandra Mathioni and David Martin for the corrections in English. Additional funds were provided to A.N. by the National Council for Scientific and Technological Development (CNPq) through a PDE scholarship (grant number 234000/2014-7).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2019.117633.

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